

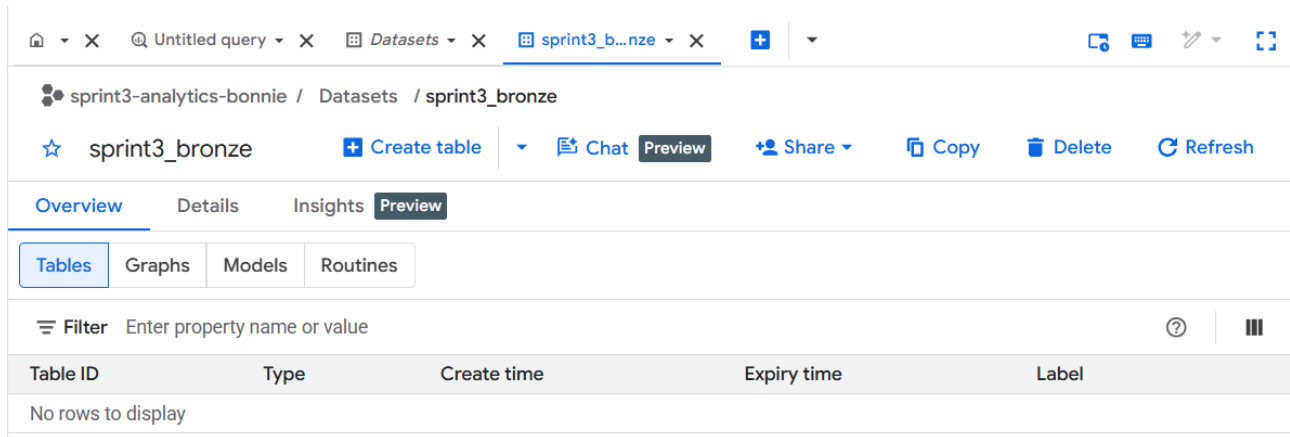
SPRINT 3 – Big Query

(P2P notes in green, feedback from Jordi Calmet)

EXERCISE 1:

Dataset Físic sprint3_bronze , mètode: Utilitza la **Interfície Gràfica (UI)** de BigQuery (fent clics).

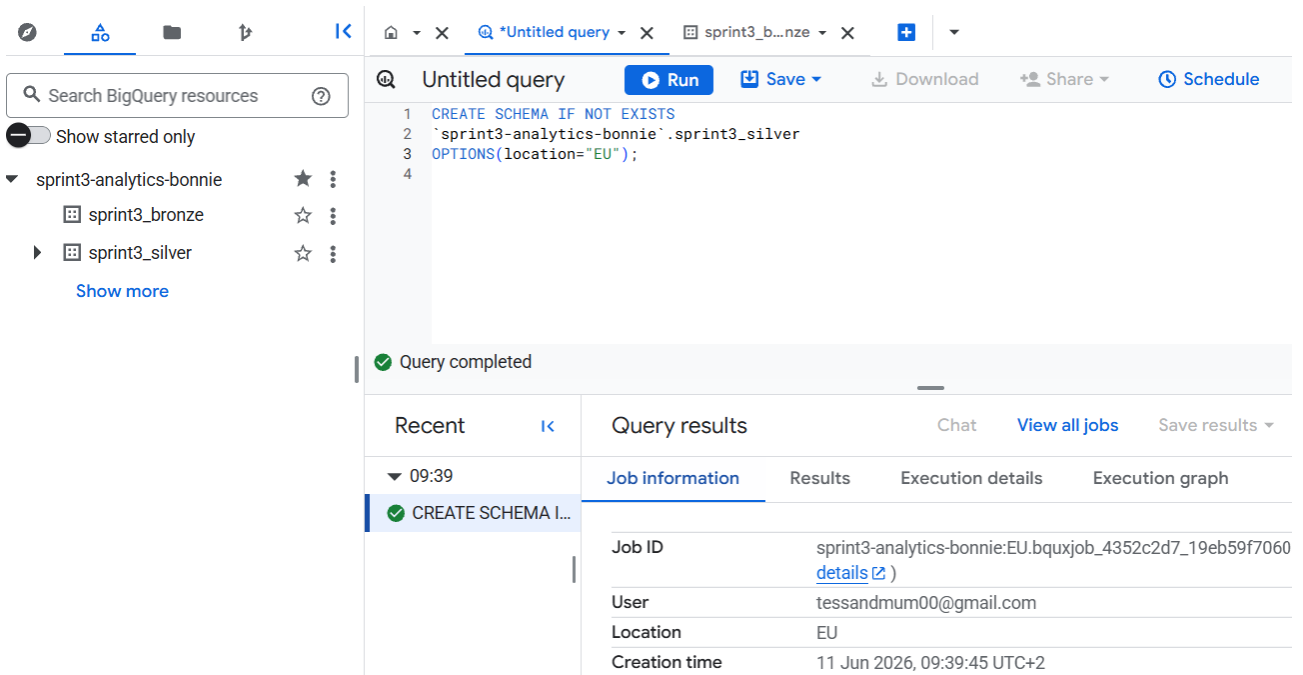
Bronze level dataset created via UI in BigQuery:



Dataset Físic sprint3_silver

- Funció: Dades netes, tipades i deduplicades.
- Mètode: Utilitza codi SQL (CREATE SCHEMA).

Silver level dataset created via SQL code in BigQuery studio console:



Dataset Físic sprint3_gold

- Funció: Dades agregades llestes per a informes i panells de control (dashboards).
- Mètode: Utilitza Cloud Shell (Línia d'ordres bq).

Gold level dataset created via Cloud Shell using the bq commandline:

```
(sprint3-analytics-bonnie) x + ▾ Editor [Icons]

i CLI.
Your Cloud Platform project in this session is set to sprint3-analytics-bonnie.
Use `gcloud config set project [PROJECT_ID]` to change to a different project.
tessandmum00@cloudshell:~ (sprint3-analytics-bonnie)$ bk mk \
> --dataset \
> -- location=EU \
> sprint3-analytics-bonnie:sprint3_gold
-bash: bk: command not found
tessandmum00@cloudshell:~ (sprint3-analytics-bonnie)$ ^C
tessandmum00@cloudshell:~ (sprint3-analytics-bonnie)$ bq mk \
> --dataset \
> -- location=EU \
> sprint3-analytics-bonnie:sprint3_gold
Dataset name: location=EU is invalid, must be letters (uppercase or lowercase), numbers, and
underscores up to 1024 characters.
tessandmum00@cloudshell:~ (sprint3-analytics-bonnie)$ ^C
tessandmum00@cloudshell:~ (sprint3-analytics-bonnie)$ bq mk \
> --dataset \
> --location=EU \
> `sprint3-analytics-bonnie`:sprint3_gold
-bash: sprint3-analytics-bonnie: command not found
Dataset 'sprint3-analytics-bonnie:sprint3_gold' successfully created.
tessandmum00@cloudshell:~ (sprint3-analytics-bonnie)$ bq ls --project_id='sprint3-analytics-bonnie'
> bq show --format=prettyjson sprint3-analytics-bonnie:sprint3_gold
> █
```

(error due to "-" in project name, resolved using back-ticks (`) to enclose the project name)

checking creation: new sprint3_gold dataset showing in project details:

Search BigQuery resources

Show starred only

Data preparations

Pipelines

Connections

sprint3_bronze

sprint3_gold

sprint3_silver

sprint3-analytics-bonnie / Datasets / sprint3_gold

☆ sprint3_gold

+ Create table

Chat

Preview

Overview

Details

Insights

Preview

Data set info

Data set ID	sprint3-analytics-bonnie.sprint3_gold
Created	11 Jun 2026, 10:27:58 UTC+2
Default table expiry	60 days
Last modified	11 Jun 2026, 10:27:58 UTC+2
Data location	EU

EXERCISE 2:

Creation of external tables for each of the csv files shared, and loading the data directly on creation:

Creation of transaction_raw table (delimiter ";")

```

5  -- EXERCISE 2
6
7  CREATE OR REPLACE EXTERNAL TABLE sprint3_bronze.transactions_raw
8  OPTIONS (
9    format = 'CSV',
10   uri = ['gs://bootcamp-data-analytics-public/ERP/transactions.csv'],
11   field_delimiter = ',';
12 );
13
14 SELECT*
15 FROM `sprint3_bronze.transactions_raw`
16 LIMIT 10;

```

✓ This query will process 0 B when run.

Recent	Query results	Chat	Vi
12:18 ✓ SELECT* FROM `spr... 12:17 ✓ CREATE OR REPLA... 09:39	Job information Results Execution details Execution graph i This statement created a new table named transactions_raw.		

Creation of companies_raw table:

Downloaded the csv file using the Google Cloud Storage web > 'go to path' tool (to open locally in order to see column names, separator etc and be able to create the schema manually):

(link to the file in Cloud Storage, accessed within my project/user:

<https://console.cloud.google.com/storage/overview;tab=overview?project=sprint3-analytics-bonnie>)

Google Cloud

sprint3-analytics-Bonnie

Search (/) for resources, docs, products and more

Search

Cloud Storage

Overview

Buckets

Monitoring

Settings

Storage Intelligence

Insights datasets

Configuration

Marketplace

Release notes

Object details

Buckets > bootcamp-data-analytics-public > ERP > companies.csv

Live object

Version history

Download

Edit metadata

Edit access

Delete

i Since you are not authorized to know the public access status of this object, it is possible that the put displayed is not valid.

Overview

Type	text/csv
Size	10.4 KB
Created	23 Nov 2025, 18:37:05
Last modified	23 Nov 2025, 18:37:05
Storage class	Standard
Custom contexts ?	—
Custom time	—
Public URL ?	https://storage.googleapis.com/bootcamp-data-analytics-public/ERP/companies.csv
Authenticated URL ?	https://storage.cloud.google.com/bootcamp-data-analytics-public/ERP/companies.csv
Cloud Storage URI ?	gs://bootcamp-data-analytics-public/ERP/companies.csv

companies_raw table created and data loaded:

The screenshot shows a SQL query editor with the following code:

```
12 );  
13  
14 CREATE OR REPLACE EXTERNAL TABLE sprint3_bronze.companies_raw  
15 (  
16   company_id STRING,  
17   company_name STRING,  
18   phone STRING,  
19   email STRING,  
20   country STRING,  
21   website STRING  
22 )  
23 OPTIONS (  
24   format = 'CSV',  
25   skip_leading_rows = 1,  
26   uris = ['gs://bootcamp-data-analytics-public/ERP/companies.csv'],  
27   field_delimiter = ','  
28 );
```

Below the code, a green checkmark icon and the text "Query completed" are displayed. Underneath, there is a section titled "Query results" with tabs for "Job information", "Results", "Execution details", and "Execution graph". The "Results" tab is selected, showing an information icon and the text: "This statement created a new table named companies_raw."

Creation of remaining tables (schemas autodetected and no special technical notes):

american_users_raw	CRM	gs://bootcamp-data-analytics-public/CRM/american_users.csv	Estàndard.
european_users_raw	CRM	gs://bootcamp-data-analytics-public/CRM/european_users.csv	Estàndard.
credit_cards_raw	CRM	gs://bootcamp-data-analytics-public/CRM/credit_cards.csv	Estàndard.

The screenshot shows a SQL query editor with the following code:

```
31 CREATE OR REPLACE EXTERNAL TABLE sprint3_bronze.american_users_raw  
32 OPTIONS (  
33   format = 'CSV',  
34   uris = ['gs://bootcamp-data-analytics-public/CRM/american_users.csv']  
35 );
```

Below the code, a green checkmark icon and the text "Query completed" are displayed. Underneath, there is a section titled "Query results" with tabs for "Job information", "Results", "Execution details", and "Execution graph". The "Results" tab is selected, showing an information icon and the text: "This statement created a new table named american_users_raw."

```

37 CREATE OR REPLACE EXTERNAL TABLE sprint3_bronze.european_users_raw
38 OPTIONS (
39   format = 'CSV',
40   uris = ['gs://bootcamp-data-analytics-public/CRM/european_users.csv']
41 );

```

✓ Query completed

>| Query results

Job information Results Execution details Execution graph

❗ This statement created a new table named european_users_raw.

```

43 CREATE OR REPLACE EXTERNAL TABLE sprint3_bronze.credit_cards_raw
44 OPTIONS (
45   format = 'CSV',
46   uris = ['gs://bootcamp-data-analytics-public/CRM/credit_cards.csv']
47 );

```

✓ Query completed

>| Query results

Job information Results Execution details Execution graph

❗ This statement created a new table named credit_cards_raw.

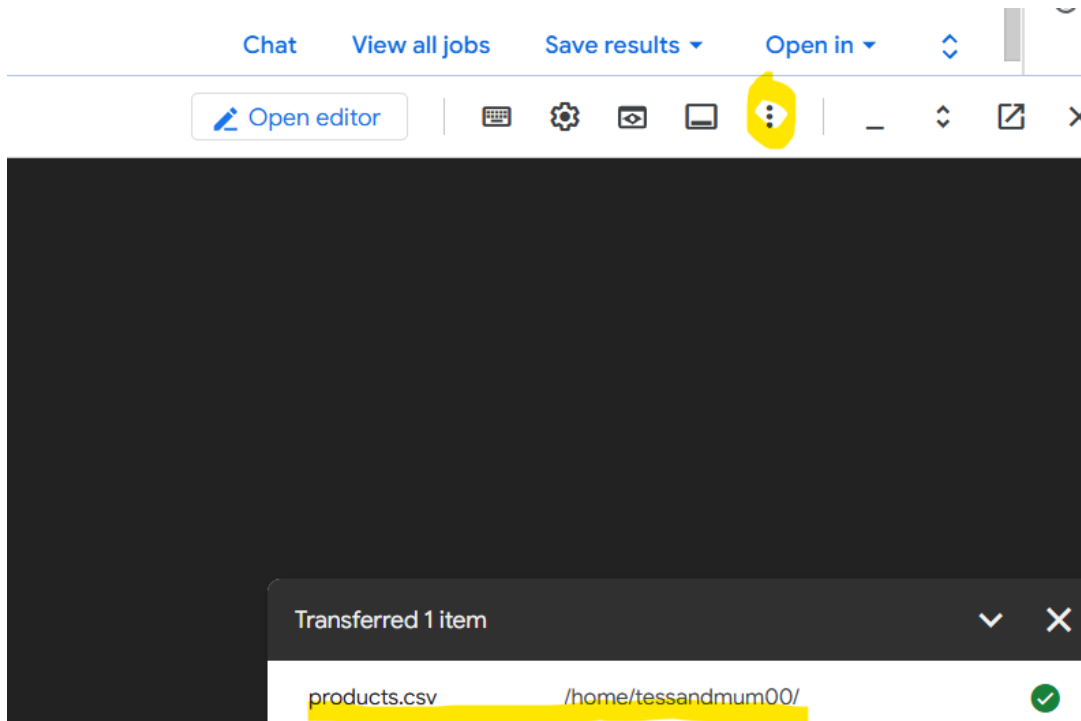
All 5 bronze level tables now created and data loaded:

- ▼  sprint3_bronze ☆ ⋮
 -  american_users_raw ☆ ⋮
 -  companies_raw ☆ ⋮
 -  credit_cards_raw ☆ ⋮
 -  european_users_raw ☆ ⋮
 -  transactions_raw ☆ ⋮
 - ▶  sprint3_gold ☆ ⋮
 - ▶  sprint3_silver ☆ ⋮

EXERCISE 3: Loading local data (upload)

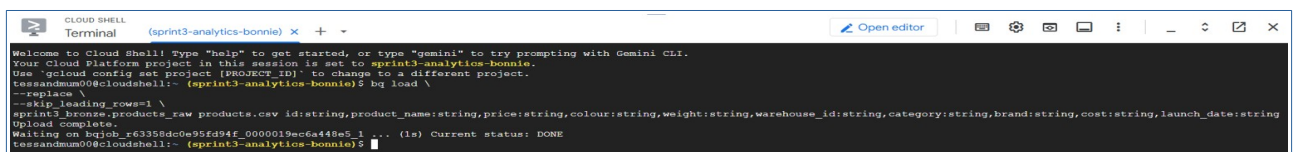
Upload the *products.csv* file from Sprint 2 as *products_raw* (will be the only native table)

1) using the bq shell, clicked MORE > UPLOAD and selected the csv file to upload it so that the shell can access it:



2) using the bq shell again to load the data from the csv file, specifying *skip_leading_row = 1*, and also the schema (column_name:string, etc), as follows:

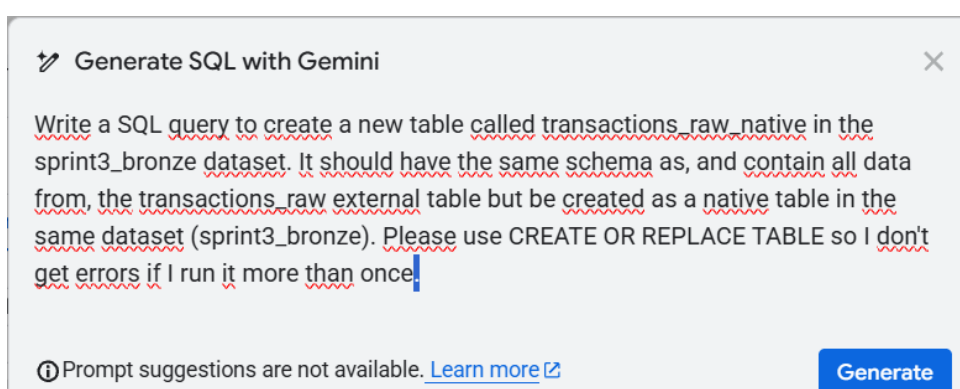
```
bq load \  
--replace \  
--skip_leading_rows=1 \  
sprint3_bronze.products_raw products.csv \  
id:string,product_name:string,price:string,colour:string,weight:string,warehouse_id:string,category:string,brand:string,cost:string,launch_date:string
```



EXERCISE 4:

Use AI assistance to get a query to make a native copy of the *transactions_raw* external table.

My prompt for Gemini:



the proposed query from Gemini:

Generate SQL with Gemini

Write a SQL query to create a new table called `transactions_raw_native` in the `sprint3_bronze` dataset. It should have the same schema as, and contain all data from, the `transactions_raw` external table but be created as a native table in the same dataset (`sprint3_bronze`). Please use `CREATE OR REPLACE TABLE` so I don't get errors if I run it more than once.

Edit

```
CREATE OR REPLACE TABLE
`sprint3-analytics-bonnie`.`sprint3_bronze`.`transactions_raw_native` AS
SELECT
  id,
  card_id,
  business_id,
  timestamp,
  amount,
  declined,
  product_ids,
  user_id,
  lat,
  longitude
FROM
`sprint3-analytics-bonnie`.`sprint3_bronze`.`transactions_raw`;
```

✓ This query will process 0 B when run.

< 1/1 >

Refine

Edit table sources Query summary [Use query with caution](#)

Rate this suggestion

Insert

I choose to **refine the script** (SELECT*) to simplify it and make more readable/explicit that it is an exact copy of the original table.

Refine

replace the list of columns in the SELECT statement with * (... AS SELECT* FROM...

I am happy refining the query this way because this is a one-time copy of the table (to carryout ad-hoc cost analysis) and not part of a pipeline that would be repeated following a schedule (where any changes to the columns in the source table could cause disruptions further down the pipeline if new columns were added at a later date, and thus would be included in my SELECT* instruction).

Refined query:

Generate SQL with Gemini

replace the list of columns in the SELECT statement with * (... AS SELECT* FROM...)

```

CREATE OR REPLACE TABLE
| `sprint3-analytics-bonnie`.`sprint3_bronze`.`transactions_raw_native` AS
SELECT
| *
FROM
| `sprint3-analytics-bonnie`.`sprint3_bronze`.`transactions_raw`;

```

✓ This query will process 0 B when run.

☐ Show diff < 2/2 >

Refine

[Edit table sources](#)
[Query summary](#)
[Use query with caution](#)

Rate this suggestion   

Insert

execution of refined query:

```

54 -- 1. Write a SQL query to create a new table called transactions_raw_native in the sprint3_bronze dataset. It should have the same schema as,
    and contain all data from, the transactions_raw external table but be created as a native table in the same dataset (sprint3_bronze). Please
    use CREATE OR REPLACE TABLE so I don't get errors if I run it more than once.
55 -- 2. replace the list of columns in the SELECT statement with * (... AS SELECT* FROM...)
56 CREATE OR REPLACE TABLE
57 | `sprint3-analytics-bonnie`.`sprint3_bronze`.`transactions_raw_native` AS
58 SELECT
59 | *
60 FROM
61 | `sprint3-analytics-bonnie`.`sprint3_bronze`.`transactions_raw`;
62

```

✓ Query completed

Recent < Query results Chat View all jobs Save results Open in

☒ CREATE OR REPL... Job information Results Execution details Execution graph

☒ CREATE OR REPL...

☒ CREATE OR REPL...

☒ CREATE OR REPL...

 This statement created a new table named transactions_raw_native. Go to table

table created (storage details show that this copy is a native table):

sprint3-analytics-bonnie / Datasets / sprint3_bronze / Tables / transactions_raw_native

☆ transactions_raw_native Query Chat Open in Share

Schema	Details	Preview	Table explorer	Preview	Insights	Lineage
Table ID	sprint3-analytics-bonnie.sprint3_bronze.transactions_raw_native					
Created	11 Jun 2026, 18:59:12 UTC+2					
Last modified	11 Jun 2026, 18:59:12 UTC+2					
Table expiry	10 Aug 2026, 18:59:12 UTC+2					
Data location	EU					
Default collation						
Default rounding mode	ROUNDING_MODE_UNSPECIFIED					
Case insensitive	false					
Description						
Labels						
Primary key(s)						
Tags						
Storage info ⓘ						
Number of rows	100,000					
Total logical bytes	10.87 MB					
Active logical bytes	10.87 MB					
Long-term logical	0 B					

where as the details for the original table that this was copied from show the external data configuration:

sprint3-analytics-bonnie / Datasets / sprint3_bronze / Tables / transactions_raw

☆ transactions_raw Query Chat Open in Share

Schema	Details	Insights	Lineage	Data profile	Data Quality
Table ID	sprint3-analytics-bonnie.sprint3_bronze.transactions_raw				
Created	11 Jun 2026, 18:59:03 UTC+2				
Last modified	11 Jun 2026, 18:59:03 UTC+2				
Table expiry	10 Aug 2026, 18:59:03 UTC+2				
Data location	EU				
Case insensitive	false				
Description					
Labels					
Primary key(s)					
Tags					
External data configuration					
Source URI(s)	gs://bootcamp-data-analytics-public/ERP/transactions.csv				
Auto-detect schema	true				
Source format	CSV				
Field delimiter (CSV)	;				

b) comparison of estimated processing costs between native and external tables:

simple query on native table:

```
63 SELECT id
64 FROM sprint3_bronze.transactions_raw_native
65 ORDER BY id;
66
```

🟢 This query will process 3.62 MB when run.

same query on external table,

```
63 SELECT id
64 FROM sprint3_bronze.transactions_raw
65 ORDER BY id;
66
```

✓ This query will process 0 B when run.

(NB: the dry run shows “0 B” as it is an external table, so cannot calculate compute until it connects with the data when it is executed).

The following test queries were executed (are not dry run estimates):

	native table (transactions_raw_native)		external table (transactions_raw)	
	<i>processed</i>	<i>billed</i>	<i>processed</i>	<i>billed</i>
SELECT id FROM sprint3_bronze.[table] ORDER BY id;	3.62 MB	10 MB	12.61 MB	13 MB
SELECT id FROM sprint3_bronze.[table];	3.62 MB	10 MB	12.61 MB	13 MB
SELECT id FROM sprint3_bronze.[table] WHERE id LIKE "%1%" ORDER BY id DESC;	3.62 MB	10 MB	12.61 MB	13 MB
SELECT COUNT(id) FROM sprint3_bronze.[table] WHERE id LIKE "%1%";	3.62 MB	10 MB	12.61 MB	13 MB

As shown by the results above of the tests run, there is a significant difference between the **bytes processed** on the native table in comparison to those processed for the external table. The native table processes less than 30% of the bytes processed in the queries on the external table.

However, this difference is not translated as significantly to the corresponding **bytes billed** for each query. The “cost” in MB of the native table query processing is only 30% less - not 70% less - than that of processing queries on the external table.

I understand that this absorption of the benefits of working with the native table is due to tiered billing charges, fixing a set billing amount for a range of bytes processed (for up to a certain amount of MB the billing “cost” is the same).

Also further studies would need to be carried out focused on the storage charges for the native table and if this outweighs the small benefit in compute costs. The sample tables used in the tests are very small (Total logical bytes 10.87 MB and Total physical bytes 4.58 MB), but on larger tables the storage costs of a native table may become significant.

In conclusion, the test carried-out is inconclusive as a general benchmark study, due to tiered billing and storage costs that would depend on specific characteristics of the data to be stored/processed. The size of the actual data tables to be used would need to be factored in to the study in order to extract true conclusions according to each specific case/project.

c) `SELECT * FROM transactions_raw_native LIMIT 10` sobre la taula **externa**.

Comprovació: Mira el "Dry Run". Ha baixat el cost respecte a no posar LIMIT? Explica breument per què això és una trampa per a principiants.

NB: the **external** table is `transactions_raw`, and not `transactions_raw_native` as stated in the suggested script in the question...

EXTERNAL TABLE:

I get an estimation of 0 bytes in the dry run banner when I point the `SELECT*` query to the external table:

```
74 SELECT*
75 FROM sprint3_bronze.transactions_raw
76
✓ This query will process 0 B when run.
```

If I add the `LIMIT 10` instruction I get the same 0 bytes.

I understand that this is because BigQuery dry run does not know the amount of processing that will be needed as it doesn't know the size of the external table until the query is run and the connection is made.

Therefore, I deduce that the test should be carried-out on the native table, `transactions_raw_native` and not the "external" table as was stated.

NATIVE TABLE:

For the same query, but directed to the native table, dry run estimates 10.87 MB of data processing:

```
74 SELECT*
75 FROM sprint3_bronze.transactions_raw_native;
76
✓ This query will process 10.87 MB when run.
```

If I add the `LIMIT 10` instruction, I get the exact same data processing estimation:

```
74 SELECT*
75 FROM sprint3_bronze.transactions_raw_native
76 LIMIT 10;
77
✓ This query will process 10.87 MB when run.
```

This is due to the fact that the standard execution order of SQL queries establishes that the `LIMIT` instruction is applied *after* the `SELECT` statement has been executed (in this case the `SELECT` is on ALL the data in the table), and it only limits the results to be shown. In other words, the processing has already been carried out before the instruction to limit is reached and applied. The limit is only applied to the results to be shown and not the amount of data that had to be processed.

On running the queries, I get the exact same MB processed and billed both with and without the `LIMIT 10`:

74 SELECT*	75 FROM sprint3_bronze.transactions_raw_native	76 LIMIT 10;	77
✓ Query completed			
Using on-demand processing quota			
Recent	Query results		
▼ 12:58	Job information	Results	Visualisation
✓ SELECT* FROM spri...	Duration	0 sec	
	Bytes processed	10.87 MB	
	Bytes billed	11 MB	
	Slot milliseconds	31	
	Job priority	INTERACTIVE	

74 SELECT*	75 FROM sprint3_bronze.transactions_raw_native;	76
✓ Query completed		
Using on-demand processing quota		
Recent	Query results	
▼ 13:00	Job information	Results
✓ SELECT* FROM spri...	Duration	1 sec
▼ 12:58	Bytes processed	10.87 MB
✓ SELECT* FROM spri...	Bytes billed	11 MB
	Slot milliseconds	1220
	Job priority	INTERACTIVE
	Use legacy SQL	false
	Destination table	Temporary table

EXERCISE 5:

Exercici 5: Adaptació de Sintaxi (Reporting)

El teu cap vol saber quins van ser els **5 dies amb més ingressos de l'any 2021**.

Repte:

Probablement el camp timestamp és un STRING. Hauràs d'investigar funcions de BigQuery (SUBSTR, CAST, PARSE_TIMESTAMP) per filtrar l'any i agrupar per data correctament

My timestamp fields, in both the native and the external bronze level transactions tables, are TIMESTAMP data type, "2020-08-20 16:55:54 UTC".

☐	timestamp	TIMESTAMP	NULLABLE
--------------------------	-----------	-----------	----------

Query executed and results obtained:

The screenshot shows a BigQuery interface. At the top, a SQL query is displayed with line numbers 80 to 85. The query is: `SELECT ROUND(SUM(amount),2) AS total_billed, DATE(timestamp) FROM `sprint3-bronze.transactions_raw_native` WHERE DATE(timestamp) BETWEEN '2021-01-01' AND '2021-12-31' GROUP BY DATE(timestamp) ORDER BY SUM(amount) DESC LIMIT 5;`. Below the query, a status bar indicates 'Query completed' and 'Using on-demand processing quota'. The main area is divided into 'Recent' and 'Query results'. The 'Query results' section shows a table with 5 rows of data. The table has columns 'Row', 'total_billed', and 'f0_'. The data is as follows:

Row	total_billed	f0_
1	11440.81	2021-10-27
2	11328.84	2021-01-22
3	11182.16	2021-02-11
4	11119.01	2021-04-29
5	10898.51	2021-03-27

P2P FEEDBACK: I didn't apply the `WHERE declined = 0` filter, and another option for filtering by the year would be:

```
WHERE EXTRACT(year FROM timestamp)
```

EXERCISE 6:

Consultes Complexes: Necessitem un informe que creui dades.

Llista el nom, país i data de les transaccions realitzades per empreses que van fer operacions entre 100 i 200 euros en alguna d'aquestes dates: 29-04-2015, 20-07-2018 o 13-03-2024.

Query executed: result shows 19 transactions carried out on these dates and with amounts between 100 and 200 euros.

```
103 SELECT co.company_name, co.country, DATE(tr.timestamp) AS transaction_date, tr.amount
104 FROM sprint3_bronze.companies_raw co
105 JOIN sprint3_bronze.transactions_raw_native tr
106 ON tr.business_id = co.company_id
107 WHERE DATE(tr.timestamp) IN ("2015-04-29", "2018-07-20", "2024-03-13")
108 AND tr.amount BETWEEN 100 and 200
109 AND declined = 0
```

Query completed

Using on-demand processing quota

Query results

Chat

View all jobs

Save results

Job information Results Visualisation JSON Execution details Execution graph

Row	company_name	country	transaction_date	amount
1	Donec Fringilla PC	France	2015-04-29	187.52
2	Netus Et Malesuada Ltd	Netherlands	2018-07-20	183.64
3	Auctor Mauris Vel LLP	United States	2015-04-29	181.42
4	Non Magna LLC	United Kingdom	2015-04-29	178.1
5	Dictum Eu Corp.	Canada	2015-04-29	177.67

Results per page: 50 1 - 19 of 19

NIVEL 2

EXERCISE 1:

Neteja de Productes (Data Quality)

Crearem la capa neta ("Silver") per als productes. Tens la

taula `sprint3_bronze.products_raw` carregada des del CSV. L'equip de Data Governance et

demana crear una taula de productes neta a `sprint3_silver.products_clean` que compleixi

aquestes regles de qualitat:

- » **Estandardització de Noms:** La columna original `id` és ambigua; reanomena-la a `product_id`. La columna `product_name` simplifica-la a `name`.
- » **Neteja d'IDs:** El camp `warehouse_id` té el format antic "WH-4". Elimina el prefix "WH-" i converteix el valor a número enter (INT64).
- » **Garantia de Preu:** Assegura't que `price` és un número (FLOAT64), sense símbols de moneda.
- » **Altres columnes:** Conserva el camp `weight` (pes) tal qual.

```
104 CREATE TABLE sprint3_silver.products_clean AS
105 SELECT
106   id AS product_id,
107   product_name AS name,
108   SAFE_CAST((REPLACE(price, '$', ''))) AS FLOAT64 AS price,
109   colour,
110   weight,
111   SAFE_CAST((REPLACE(warehouse_id, 'WH-', ''))) AS INT64 AS warehouse_id,
112   category,
113   brand,
114   SAFE_CAST((REPLACE(cost, '$', ''))) AS FLOAT64 AS cost,
115   SAFE_CAST(launch_date AS DATE) AS launch_date
116 FROM sprint3_bronze.products_raw;
```

Query completed

Using on-demand processing quota

Query results

Chat

Job information Results Execution details Execution graph

This statement created a new table named `products_clean`.

P2P FEEDBACK:

Here I didn't detect that some of the original values in warehouse_id had double "-", which although have converted to INT64 without producing an error when my script was executed, they will have registered as negative numerical values.

Options explained to remedy this:

- Using a regular expression:
`SAFE_CAST(REGEXP_REPLACE(warehouse_id AS STRING), r'^0-9', '') AS INT64) AS warehouse_id`
- Using an ABS(), to have only absolute values:
`SAFE_CAST(ABS(REPLACE(warehouse_id, 'WH-', '')) AS INT64) AS warehouse_id`

EXERCISE 2:

Creació de Transaccions Netes (Capa Silver)

Escenari:

La taula transactions_raw (Bronze) és insegura: l'import (amount) podria tenir lletres, les coordenades podrien fallar i la data és un text difícil de filtrar. Crearem la taula definitiva sprint3_silver.transactions_clean que garanteixi el següent.

- » **Estandardització de Noms:** La columna original id és ambigua; reanomena-la a transaction_id.
- » **Robustesa en Imports:** Utilitza SAFE_CAST al camp amount. Si falla, substitueix-lo per 0 (usant IFNULL).
- » **Dates Reals:** Converteix el camp timestamp (que és STRING) a tipus TIMESTAMP real.
- » **Coordenades:** Assegura que lat i longitude siguin FLOAT64 (utilitza SAFE_CAST per seguretat).
- » **Desgloss de Productes:** Transforma la cadena de text product_ids en un **ARRAY de enteros**.
- *Input:* "1, 2, 3" (String)
- *Output:* [1, 2, 3] (Array/Lista de enteros)
- » **Resta de camps:** Mantén-los igual.

NB:

In my bronze level transactions_raw table (schema automatically established on importing data) :

- the amount field is data type FLOAT, doesn't contain letters or NULL values or value 0.
- the timestamp field is already data type TIMESTAMP.
- The coordinates are already FLOAT data type.

☆	transactions_raw	Query	Chat	Op
Schema	Details	Insights	Lineage	Data profile
Filter Enter property name or value				
☐	Field name	Type	Mode	Des
☐	id	STRING	NULLABLE	-
☐	card_id	STRING	NULLABLE	-
☐	business_id	STRING	NULLABLE	-
☐	timestamp	TIMESTAMP	NULLABLE	-
☐	amount	FLOAT	NULLABLE	-
☐	declined	INTEGER	NULLABLE	-
☐	product_ids	STRING	NULLABLE	-
☐	user_id	INTEGER	NULLABLE	-
☐	lat	FLOAT	NULLABLE	-
☐	longitude	FLOAT	NULLABLE	-

Table created and populated:

143 CREATE OR REPLACE TABLE sprint3_silver.transactions_clean AS
144 SELECT
145 id AS transaction_id,
146 card_id,
147 business_id,
148 SAFE_CAST(timestamp AS TIMESTAMP) AS timestamp,
149 SAFE_CAST(amount AS FLOAT64) AS amount,
150 declined,
151 ARRAY(SELECT CAST(products AS INT64)
152 FROM UNNEST(SPLIT(product_ids, ',')) AS products
153) AS product_ids,
154 user_id,
155 SAFE_CAST(lat AS FLOAT64) as lat,
156 SAFE_CAST(longitude AS FLOAT64) as longitude

✔ NOTE: Could not compute bytes processed estimate for script.

Recent1K

Query resultsCt

▼ 08:38

▼ 08:22

Job informationResultsExecution detailsExecution graph

✔ CREATE OR REPL...

ⓘ This statement created a new table named transactions_clean.

preview of new table:

sprint3-analytics-bonnie / Datasets / sprint3_silver / Tables / transactions_clean

☆ transactions_... 🔍 Query 🗨 Chat ⌵ Open in ▾ 👤 Share ▾ 📄 Copy 📷 Snapshot 🗑 Delete ⬆ Export ↻ Refresh

SchemaDetailsPreviewTable explorerPreviewInsightsLineageData profileData Quality

	transaction_id	card_id	business_id	timestamp	amount	declined	product_ids
1	26904E68-80E1-41DF-9ECB-3A...	CcS-4857	b-2406	2016-10-12 12:27:29 UTC	6.9	0	95
2	621CB9D6-7896-401D-BD38-3F...	CcS-4857	b-2402	2018-03-02 23:33:26 UTC	38.33	0	98
3	BDD988A9-51B9-4BD8-9E4C-84...	CcS-4857	b-2610	2024-10-25 13:11:54 UTC	129.3	0	46
							48
							24
4	6BE556F4-9B3F-4C64-A4E2-23...	CcS-4857	b-2442	2022-05-20 08:36:18 UTC	639.02	0	14
							39
							65
							27
5	CA32CD20-B2CC-4BA7-8B57-9...	CcS-4857	b-2442	2021-10-02 03:58:28 UTC	376.99	0	40
							59
							45
							87
6	B954A0B3-9314-4615-ACBB-B3...	CcS-4857	b-2558	2024-10-07 16:43:17 UTC	107.17	0	35

Results per page: 50 ▾ 1 - 50 of 100000 |< > ⌵

P2P FEEDBACK: I didn't apply the IFNULL fix on the amount:

```
IFNULL(SAFE_CAST(amount AS FLOAT64), 0) AS amount
```

EXERCISE 3:

Unificació d'Usuaris (UNION)

Tenim els usuaris fragmentats per regió.

Crea la taula `sprint3_silver.users_combined`. Utilitza `UNION ALL` per unificar els usuaris dels EUA i Europa en una única llista mestra. Afegeix una columna calculada `origin` per saber d'on venen.

- » **Estandardització de Noms:** La columna original `id` és ambigua; reanomena-la a `user_id`.

Table created and data from the 2 raw user tables incorporated:

```
72 CREATE OR REPLACE TABLE sprint3_silver.users_combined AS
173 SELECT
174     "American" as origin,
175     id AS user_id,
176     name,
177     surname,
178     phone,
179     email,
180     PARSE_DATE('%b %e, %Y', birth_date) AS birth_date,
181     country,
182     city,
183     postal_code,
184     address
185 FROM sprint3_bronze.american_users_raw
186 UNION ALL
187 SELECT
188     "European" as origin,
189     id AS user_id,
190     name,
191     surname,
192     phone,
193     email,
194     PARSE_DATE('%b %e, %Y', birth_date) AS birth_date,
```

Query completed

Recent	Query results
11:43	Job information Results Execution details Execution graph
CREATE OR REPL...	This statement created a new table named users_combined.

full code (cut-off in screenshot):

```
CREATE OR REPLACE TABLE sprint3_silver.users_combined AS
SELECT
    "American" as origin,
    id AS user_id,
    name,
    surname,
    phone,
    email,
    PARSE_DATE('%b %e, %Y', birth_date) AS birth_date,
    country,
    city,
    postal_code,
    address
FROM sprint3_bronze.american_users_raw
UNION ALL
SELECT
    "European" as origin,
    id AS user_id,
    name,
    surname,
    phone,
    email,
    PARSE_DATE('%b %e, %Y', birth_date) AS birth_date,
```



```

country,
city,
postal_code,
address
FROM sprint3_bronze.european_users_raw;

```

Preview of table created and populated with the 5000 records from the 2 bronze level tables:

sprint3-analytics-bonnie / Datasets / sprint3_silver / Tables / users_combined

☆ users_combined Query Chat Open in ▾ Share ▾ Copy Snapshot Delete Export Refresh

Schema Details **Preview** Table explorer Preview Insights Lineage Data profile Data Quality

Row	origin	user_id	name	surname	phone	email	birth_date	country	city	postal_code	address
1	American	1497	Etcuryh	Irysyzge	+55-760-9477	etcuyh.iryzyzge@example.com	2005-03-18	Canada	Quebec City	G1A 0A1	335 Irysyzge St
2	American	2297	Zkczde	Caquustp	+77-662-6909	zkczde.caquustp@example.com	1998-01-14	Canada	Quebec City	G1A 0A1	326 Caquustp St
3	American	4058	Rsigkq	Mhmabmpe	+63-952-1552	rsigkq.mhmabmpe@example.c...	1973-02-11	Canada	Quebec City	G1A 0A1	140 Mhmabmpe St
4	American	203	Jarrod	Fields	010-741-8105	sit.amet@google.couk	1982-01-06	Canada	Quebec City	G1A 0A1	441-8969 Rhoncus R
5	American	1401	Devnbq	Zlqgskuu	+71-537-6029	devnbq.zlqgskuu@example.com	1950-09-22	Canada	Quebec City	G1A 0A1	950 Zlqgskuu St
6	American	1405	Chjqfw	Pzbfjhvp	+61-591-8811	chjqfw.pzbfjhvp@example.com	1975-07-24	Canada	Quebec City	G1A 0A1	242 Pzbfjhvp St
7	American	2013	Erswn	Ohijzach	+75-206-6920	erswn.ohijzach@example.com	1975-11-08	Canada	Quebec City	G1A 0A1	11 Ohijzach St
8	American	4438	Wypaxn	Abwzprww	+40-290-2892	wypaxn.abwzprww@example.c...	1992-09-08	Canada	Quebec City	G1A 0A1	248 Abwzprww St
9	American	1772	Rysxpq	Jkfeheuh	+45-547-8984	rysxpq.jkfeheuh@example.com	2002-08-21	Canada	Quebec City	G1A 0A1	83 Jkfeheuh St
10	American	3808	Blmbve	Dcpfrpgf	+57-227-9971	blmbve.dcpfrpgf@example.com	1958-06-21	Canada	Quebec City	G1A 0A1	501 Dcpfrpgf St
11	American	238	Chester	Haynes	059-783-3104	ut@protonmail.edu	1988-09-13	Canada	Quebec City	G1A 0A1	P.O. Box 284
12	American	1284	Hhbred	Extswwhr	+74-147-2392	hhbred.extswwhr@example.com	1975-03-14	Canada	Quebec City	G1A 0A1	375 Extswwhr St
13	American	1760	Gmhw...	Ioxltfyv	+56-494-8132	gmhwrp.ioxltfyv@example.com	1986-02-14	Canada	Quebec City	G1A 0A1	916 Ioxltfyv St
14	American	3739	Dfcwtz	Gnsombfd	+92-981-9785	dfcwtz.gnsombfd@example.com	1973-06-20	Canada	Quebec City	G1A 0A1	352 Gnsombfd St
15	American	3810	Kfahfj	Zfsnhvyc	+64-488-5469	kfahfj.zfsnhvyc@example.com	2000-04-14	Canada	Quebec City	G1A 0A1	957 Zfsnhvyc St
16	American	3932	Amhiax	Irmqozry	+39-894-1178	amhiax.irmqozry@example.com	1986-07-06	Canada	Quebec City	G1A 0A1	645 Irmqozry St

Results per page: 50 ▾ 1 - 50 of 5000 < > >>

Silver level companies_clean table created and populated (native table):

```

201 CREATE OR REPLACE TABLE sprint3_silver.companies_clean AS
202 SELECT
203     company_id,
204     company_name,
205     phone,
206     email,
207     country,
208     website,
209 FROM sprint3_bronze.companies_raw;
210

```

Query completed

Recent	Query results	Chat
07:52 CREATE OR REPLA... This statement created a new table named companies_clean.	Job information Results Execution details Execution graph	

Silver level credit_cards_clean table created and populated (native table):

```

210
211 CREATE OR REPLACE TABLE sprint3_silver.credit_cards_clean AS
212 SELECT
213   id AS credit_card_id,
214   user_id,
215   iban,
216   pan,
217   pin,
218   cvv,
219   track1,
220   track2,
221   expiring_date
222 FROM sprint3_bronze.credit_cards_raw;
223

```

Query completed

Recent	Query results	Chat
07:59	Job information	Results
07:52	Execution details	Execution graph
CREATE OR REPLA...	This statement created a new table named credit_cards_clean.	

P2P FEEDBACK: I missed assuring the format of the PIN and CVV variables so that preceding 0 are not lost (PIN should be 4 digits in length and CVV should be 3 digits long). This was also explained in my SQL test yesterday with Irene...

Better as either data type STRING (assuring zeros) or CHAR (assuring exact length).

Although Primary and Foreign Keys in Big query are not enforced (i.e. even if company_id and business_id were established as PK and FK respectively, on the companies and transactions tables, I would still be able to enter a transaction record for a company even though it didn't have a record in the companies table), I have still decided to add the constraints, as they may help with the optimization of queries using JOINs (and thus cost optimization).

TABLE	KEYS
sprint3_silver.companies_clean	PK company_id
sprint3_silver.credit_cards_clean	PK credit_card_id
sprint3_silver.products_clean	PK product_id (already established)
sprint3_silver.transactions_clean	PK transaction_id FK credit_card_id FK user_id NB: unable to establish FK on product_ids as the type ARRAY is not compatible
sprint3_silver.users_combined	PK user_id

Constraints added:

```

231 # adding constraints:
232 ALTER TABLE sprint3_silver.companies_clean
233 ADD PRIMARY KEY (company_id) NOT ENFORCED;
234
235 ALTER TABLE sprint3_silver.credit_cards_clean
236 ADD PRIMARY KEY (credit_card_id) NOT ENFORCED;
237
238 ALTER TABLE sprint3_silver.users_combined
239 ADD PRIMARY KEY (user_id) NOT ENFORCED;
240
241 ALTER TABLE sprint3_silver.transactions_clean
242 ADD PRIMARY KEY (transaction_id) NOT ENFORCED,
243 ADD CONSTRAINT fk_trans_credit_cards FOREIGN KEY (card_id) REFERENCES `sprint3_silver.credit_cards_clean` (credit_card_id) NOT ENFORCED,
244 ADD CONSTRAINT fk_trans_companies FOREIGN KEY (business_id) REFERENCES `sprint3_silver.companies_clean` (company_id) NOT ENFORCED,
245 ADD CONSTRAINT fk_trans_users FOREIGN KEY (user_id) REFERENCES `sprint3_silver.users_combined` (user_id) NOT ENFORCED;

```

Query completed

Recent	Query results	Chat	View all jobs	Save results	Open in
09:21	Job information	Results	Execution details	Execution graph	
ALTER TABLE spr...	This statement altered the table named transactions_clean.				
09:21					
ALTER TABLE spr...					

view of all constraints now in place on silver level schema:

227 -- to see if any constraints in place:

228 SELECT *

229 FROM `sprint3-analytics-bonnie`.sprint3_silver.INFORMATION_SCHEMA.TABLE_CONSTRAINTS;

230

Query completed

Using on-demand processing quota

Recent

Query results

Chat

View all jobs

Save results

Open

09:24

SELECT * FROM `...

09:21

ALTER TABLE spr...

09:21

ALTER TABLE spr...

09:20

ALTER TABLE spr...

09:20

ALTER TABLE spr...

08:10

Job information

Results

Visualisation

JSON

Execution details

Execution graph

Row	constraint_catalog	constraint_schema	constraint_name	table_catalog
1	sprint3-analytics-bonnie	sprint3_silver	products_clean.pk\$	sprint3-analytics-bonnie
2	sprint3-analytics-bonnie	sprint3_silver	credit_cards_clean.pk\$	sprint3-analytics-bonnie
3	sprint3-analytics-bonnie	sprint3_silver	users_combined.pk\$	sprint3-analytics-bonnie
4	sprint3-analytics-bonnie	sprint3_silver	transactions_clean.pk\$	sprint3-analytics-bonnie
5	sprint3-analytics-bonnie	sprint3_silver	transactions_clean.fk_trans_credit_cards	sprint3-analytics-bonnie
6	sprint3-analytics-bonnie	sprint3_silver	transactions_clean.fk_trans_companies	sprint3-analytics-bonnie
7	sprint3-analytics-bonnie	sprint3_silver	transactions_clean.fk_trans_users	sprint3-analytics-bonnie
8	sprint3-analytics-bonnie	sprint3_silver	companies_clean.pk\$	sprint3-analytics-bonnie

Results per page: 50

1 - 8 of 8

NIVEL 3

EXERCISE 1:

La Vista de Màrqueting (Lògica de Negoci)

Crea una vista anomenada `sprint3_gold.v_marketing_kpis` que mostri la següent informació per a cada companyia:

- » **Nom de la companyia, Telèfon i País** (origen: `companies_clean`).
- » **Mitjana de compra** (AVG(amount) de `transactions_clean`).
- » **Classificació de Client (Lògica):**
 - Crea una columna calculada anomenada `client_tier`.
 - Si la mitjana de compra és superior a **260€**, etiqueta com a **"Premium"**.
 - Si és igual o inferior, etiqueta com a **"Standard"**.

```

259 # EXERCISE 1: vista marketing:
260
261 CREATE OR REPLACE VIEW sprint3_gold.v_marketing_kpis AS
262 SELECT co.company_name,
263        co.phone,
264        co.country,
265        av.average_spend,
266        IF(av.average_spend > 260, 'Premium', 'Standard') AS client_tier
267 FROM `sprint3_silver.companies_clean` co
268 LEFT JOIN (SELECT business_id, ROUND(AVG(amount), 2) AS average_spend
269           FROM `sprint3_silver.transactions_clean`
270           WHERE declined = 0
271           GROUP BY business_id) AS av
272 ON av.business_id = co.company_id;
273

```

✓ Query completed

Recent	Query results	Chat
06:14	Job information	Results
✓ CREATE OR REPL...	Execution details	Execution graph
ⓘ This statement replaced the view named v_marketing_kpis.		

I use a subquery in the JOIN due to the need to GROUP BY for the aggregate calculation and for readability. I edited my original VIEW ("this statement replaced...") so that:

- it uses LEFT JOIN. Now all companies are shown, including those without a transaction as yet.
- WHERE declined = 0. The average is calculated on actual completed transactions.
- Uses IF() instead of CASE as originally written and removed the ORDER BY from the view.

P2P FEEDBACK: The subquery was not necessary, can be done with a JOIN to the transactions table and then:

```
ROUND(AVG(t.amount), 2) AS avg_amount (no GROUP BY required).
```

Entrega:

Realitza una consulta SELECT * sobre la teva nova vista, ordenant els resultats perquè apareguin primer els clients "Premium" i, dins d'aquests, els que tinguin major mitjana de compra.

273	
74	SELECT*
275	FROM `sprint3-analytics-bonnie.sprint3_gold.v_marketing_kpis`
276	ORDER BY client_tier, average_spend DESC;
277	
Query completed	
Using on-demand processing quota	
Recent	Query results
▼ 06:18	Job information Results Visualisation JSON Execution details Execution graph
✓ SELECT* FROM `spr...	Row company_name phone country average_spend client_tier
▼ 06:14	1 Ac Fermentum Incorporated 06 85 56 52 33 Germany 284.91 Premium
✓ CREATE OR REPLA...	2 Pretium Neque Corp. 07 77 48 55 28 Australia 275.58 Premium
▼ 06:10	3 Urna Convallis Associates 06 01 24 77 04 United States 273.57 Premium
✓ SELECT co.compan...	4 At Associates 09 56 61 10 65 New Zealand 272.74 Premium
	5 Metus Vitae Associates 08 25 44 40 66 Australia 270.05 Premium
	6 Aliquet Diam Limited 02 76 61 47 46 United States 269.29 Premium
	7 Nec Luctus LLC 02 14 71 75 73 Norway 268.6 Premium
	8 Neque Tellus Incorporated 04 43 18 34 19 Ireland 267.56 Premium
	9 Cras Consulting 07 50 10 85 63 Belgium 267.38 Premium
	10 Sed LLC 01 63 16 26 52 Belgium 266.61 Premium
Results per page: 50 1 – 50 of 100	

EXERCISE 2:

Crea la taula `sprint3_gold.product_sales_ranking` que contingui l'inventari complet de productes i quantes vegades s'ha venut cadascun.



Requisits de l'Informe:

- » **Detall del Producte:** Ha d'incloure `product_id`, `name`, `price` i `color` (vénen de la taula `products_clean`).
- » **Mètrica de Negoci:** Una columna nova `total_sold` que compti quantes vegades apareix aquest producte a les transaccions.
- » **Integritat:** Han d'aparèixer tots els productes, fins i tot els que tenen 0 vendes (potser cal descatallar-los).



Pista Tècnica:

- » A `transactions_clean`, els productes estan "encapsulats" dins d'un Array per cada transacció.
- » Primer necessitaràs "aplanar" la taula de transaccions usant `UNNEST(product_ids)` per poder comptar els productes individualment.
- » Després, fes un encreuament (`LEFT JOIN`) començant per la taula de productes per no perdre aquells que no s'han venut mai.

Final table created:

```

299 # creating final table:
300 CREATE OR REPLACE TABLE sprint3_gold.product_sales_ranking AS
301   SELECT product_id, name, price, colour, unpacked_products.sold_count AS total_sold
302   FROM `sprint3_silver.products_clean` prod
303   LEFT JOIN (SELECT CAST(individual_product AS STRING) AS product, COUNT(individual_product) AS sold_count
304             FROM sprint3_silver.transactions_clean
305             INNER JOIN UNNEST(transactions_clean.product_ids) AS individual_product
306             GROUP BY individual_product) AS unpacked_products
307   ON unpacked_products.product = prod.product_id
308   ORDER BY unpacked_products.sold_count DESC;
309
310

```

Query completed

Using on-demand processing quota

Recent
Query results
Chat
View all jobs

14:32
Job information
Results
Execution details
Execution graph

CREATE OR REPL...

This statement created a new table named product_sales_ranking.

```

316
317 SELECT*
318 FROM `sprint3_gold.product_sales_ranking`
319 ORDER BY product_id DESC;
320

```

Query completed

Using on-demand processing quota

Query results
Chat
View all jobs
Save results
Open in

Job information
Results
Visualisation
JSON
Execution details
Execution graph

Row	product_id	name	price	colour	total_sold
1	99	the duel	151.78	#212121	2514
2	98	Direwolf Littlefinger	38.33	#bababa	2562
3	97	jinn Winterfell	65.25	#bababa	2536
4	96	dooku solo	20.92	#282828	2506
5	95	riverlands north Direwolf	6.9	#c1c1c1	2525
6	94	in south Winterfell	51.94	#3a3a3a	2479
7	93	Littlefinger Tarly Stark	192.6	#efefef	2535

Results per page: 50 1 - 50 of 100

EXERCISE 3: export the created table to csv.

Queried the new table with SELECT*, and then used: [save results > Local download, CSV] to save a copy locally.

try – you won't be charged if you run out of credit. [Learn more](#)

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Full BigQuery experience. [Learn more](#)

ex1_to_3_SPRINT3 2026-06-11 ... [Run](#) [Save](#) [Schedule](#) [Share](#)

File Edit Run Tools

```

314 ON unpacked_products.product = prod.product_id
315 ORDER BY unpacked_products.sold_count DESC;
316
317 SELECT*
318 FROM `sprint3_gold.product_sales_ranking`
319 ORDER BY product_id;

```

Query completed

Using on-demand processing quota

Query results [Chat](#) [View all jobs](#)

Job information Results Visualisation JSON Execution details Execution graph

Row	product_id	name	price	colour	total_sold
1	1	Direwolf Stannis	161.11	#7c7c7c	2468
2	10	Karstark Dorne	119.52	#f4f4f4	2571
3	100	south duel	40.43	#6d6d6d	2517
4	11	Karstark Dorne	49.7	#141414	2573
5	12	duel Direwolf	181.6	#a8a8a8	2558
6	13	palpatine chewbacca	139.59	#2b2b2b	2560
7	14	Direwolf	147.53	#c4c4c4	2496

Results per page: 50 1 - 50 of 100

Copy to Clipboard up to 1 MB

Google Drive

CSV up to 1 GB

JSONL (newline delimited) up to 1 GB

Google Sheets up to 10 MB

Local download

CSV up to 10 MB

JSON up to 10 MB

Google Cloud

Cloud Storage

BigQuery table

The downloaded CSV opened in LibreOffice Calc.

product_sales_ranking.csv • LibreOffice Calc

Archivo Editar Ver Insertar Formato Estilos Hoja Datos Herramientas Ventana Ayuda

Liberation Sans 10 pt

A1 $\sum$ product_id

	A	B	C	D	E	F	G	H
1	product_id	name	price	colour	total_sold			
2	1	Direwolf Stannis	161.11	#7c7c7c	2468			
3	10	Karstark Dorne	119.52	#f4f4f4	2571			
4	100	south duel	40.43	#6d6d6d	2517			
5	11	Karstark Dorne	49.7	#141414	2573			
6	12	duel Direwolf	181.6	#a8a8a8	2558			
7	13	palpatine chewbacca	139.59	#2b2b2b	2560			
8	14	Direwolf	147.53	#c4c4c4	2496			
9	15	Stannis warden	194.29	#dbdbdb	2535			
10	16	the duel warden	180.91	#666666	2608			
11	17	skywalker ewok sith	91.89	#7c7c7c	2527			
12	18	Karstark warden	148.91	#c4c4c4	2552			
13	19	dooku solo	60.33	#3f3f3f	2449			
14	2	Tarly Stark	9.24	#919191	2573			
15	20	warden Karstark	91.96	#b5b5b5	2497			
16	21	duel Direwolf	96.9	#e2e2e2	2609			
17	22	chewbacca mustafar	150.02	#fcfcfc	2448			
18	23	riverlands north	169.96	#545454	2593			
19	24	south duel tourney	48.99	#aaaaaa	2574			
20	25	skywalker ewok	157.53	#2b2b2b	2521			
21	26	Stark Karstark	53.01	#898989	2536			
22	27	Stannis riverlands	172.93	#7a7a7a	2503			
23	28	chewbacca mustafar	127.44	#efefef	2584			
24	29	Tully maester Tarly	167.2	#111111	2635			
25	3	duel tourney Lannister	171.13	#d8d8d8	2528			
26	30	Karstark warden	79.53	#606060	2468			
27	31	Lannister	85.02	#3f3f3f	2580			
28	32	north	178.28	#0f0f0f	2561			
29	33	duel warden	127.09	#191919	2597			
30	34	of north	128.48	#919191	2580			
31	35	the giantsblood maester	107.17	#8c8c8c	2522			
32	36	Lannister	96.22	#a5a5a5	2575			
33	37	Direwolf Littlefinger	26.66	#aaaaaa	2461			
34	38	ewok sith	55.43	#666666	2578			
35	39	warden	196.65	#a0a0a0	2571			
36	4	warden south duel	71.89	#111111	2584			
37	40	tatooine dooku solo	73.31	#070707	2488			
38	41	Lannister Barratheon Direwolf	141.01	#fcfcfc	2517			
39	42	sith	83.51	#494949	2425			

product_sales_ranking

Hoja 1 de 1 Predeterminado

Quality check:

Both the original gold-level table, product_sales_ranking, and the exported CSV file have 100 records.

(NB: as the product_id is STRING, it doesn't order numerically)